PROJECT REPORT

Heart Disease Analysis using Tableau with Flask Web Integration

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1. PROJECT TITLE

Heart Disease Analysis and Visualization using Tableau Dashboards with Flask Web Deployment

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2. PROJECT DESCRIPTION

● System Overview

This project is a data analytics web application that analyzes heart disease related data and presents insights through interactive Tableau dashboards and storytelling visualizations embedded within a Flask-based web interface.

● Problem Domain

Heart disease is one of the leading causes of mortality worldwide. Healthcare datasets are often complex and difficult to interpret. There is a need to transform raw medical data into meaningful visual insights that can support awareness and decision-making.

● Core Technology Used

- Tableau (Data Visualization & Storytelling)
- Python Flask (Web Deployment)
- HTML/CSS (User Interface)

● Key Capabilities

- Interactive visualization of heart disease risk factors
- Step-by-step data storytelling
- Web-based dashboard accessibility
- Real-time user interaction with charts

● Target Users

- Healthcare students
- Researchers
- Policy makers
- General public

● Real-world Relevance

The project helps in understanding lifestyle and demographic factors affecting heart disease and demonstrates real-world application of data analytics in healthcare.

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3. APPLICATION SCENARIOS / USE CASES

● Educational Usage
Used in classrooms to teach data visualization and healthcare analytics.

● Healthcare Awareness
Hospitals and NGOs can use dashboards to spread awareness.

● Research Usage
Researchers can explore patterns and correlations.

● End-user Usage
General users can understand personal risk factors.

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4. TECHNICAL ARCHITECTURE OVERVIEW

High Level Architecture

User → Flask Web App → Embedded Tableau Dashboard → Visualization Output

Component Interaction

- User interacts with web interface
- Flask routes serve HTML pages
- Pages contain Tableau iframe
- Tableau server renders visualizations

Data Flow

Dataset → Tableau Processing → Dashboard → Embedded in Flask → User View

Layers

Frontend Layer
HTML templates with navigation UI

Backend Layer
Python Flask routing system

Database Layer
Not applicable (static dataset used in Tableau)

External Services
Tableau Public hosting

Deployment Environment
Localhost / Web browser

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5. PREREQUISITES

5.1 Software Requirements

- Tableau Desktop / Tableau Public
- Python 3.x
- VS Code / PyCharm
- Web Browser

5.2 Libraries / Dependencies

- Flask
- pandas (optional preprocessing)

5.3 Hardware Requirements

- Minimum 8 GB RAM
- Modern browser
- Standard laptop/desktop

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6. PRIOR KNOWLEDGE REQUIRED

- Basic Python programming
- Flask routing concepts
- Tableau dashboard creation
- Data visualization fundamentals
- HTML basics

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7. PROJECT OBJECTIVES

Technical Objectives

- Build interactive dashboards
- Integrate Tableau with Flask

Performance Objectives

- Fast dashboard loading
- Smooth navigation

Deployment Objectives

- Web accessibility without Tableau software

Learning Outcomes

- Data storytelling skills
- Web deployment understanding

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8. SYSTEM WORKFLOW

1. User opens website
2. Selects Dashboard or Story
3. Flask serves HTML page
4. Tableau iframe loads visualization
5. User interacts with charts
6. Insights are observed

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9. MILESTONE 1: REQUIREMENT ANALYSIS & SYSTEM DESIGN

Problem Definition

Understanding heart disease risk patterns.

Functional Requirements

- Show dashboards
- Show story
- Provide navigation

Non-Functional Requirements

- Responsiveness
- Usability

Technology Stack Justification

Tableau provides strong visualization capability.
Flask enables lightweight deployment.

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10. MILESTONE 2: ENVIRONMENT SETUP

- Install Python
- Install Flask
- Create project folder
- Configure templates

Project Structure Created

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11. MILESTONE 3: CORE DEVELOPMENT

Feature 1
Dashboard creation in Tableau

Feature 2
Story creation

Feature 3
Flask UI integration

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12. MILESTONE 4: INTEGRATION

- Embedded iframe integration
- UI optimization
- Route validation

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13. MILESTONE 5: TESTING

Test Case Example

Test Case	Input	Expected	Status
Open Dashboard	Click tab	Dashboard	Pass

Performance Metrics

- Page load time < 3 sec

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14. DEPLOYMENT

Tableau Public hosting
Flask local deployment

Steps

- Publish dashboard
- Copy embed code
- Run Flask server

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15. PROJECT STRUCTURE

```
project_root
│
templates
static
app.py
```

templates → HTML pages
static → CSS/images
app.py → Main application

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16. RESULTS

- Successful visualization
- Interactive insights
- Smooth UI navigation

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17. ADVANTAGES & LIMITATIONS

Advantages

- Easy understanding of data
- Web accessibility
- Interactive storytelling

Limitations

- Static dataset
- Depends on Tableau Public internet

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18. FUTURE ENHANCEMENTS

- Real-time data integration
- Cloud deployment
- Mobile responsive UI
- ML prediction model

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19. CONCLUSION

This project successfully demonstrates how healthcare data can be transformed into actionable insights using visualization tools and deployed through web technologies. It enhances awareness and showcases practical implementation of data analytics.

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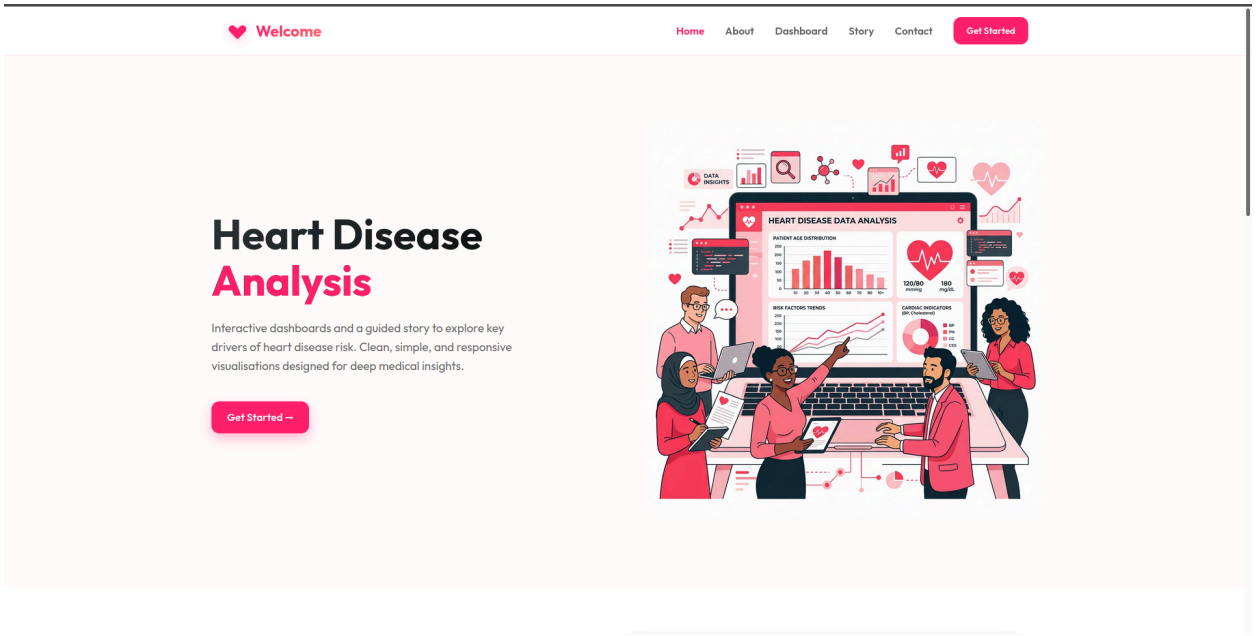
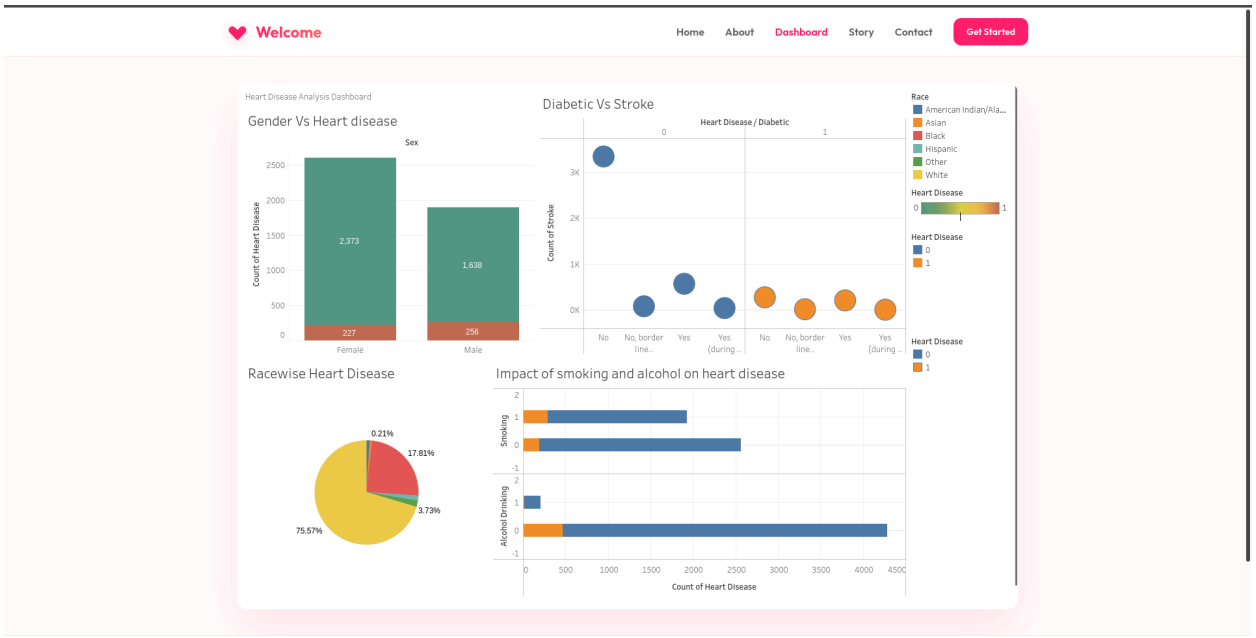
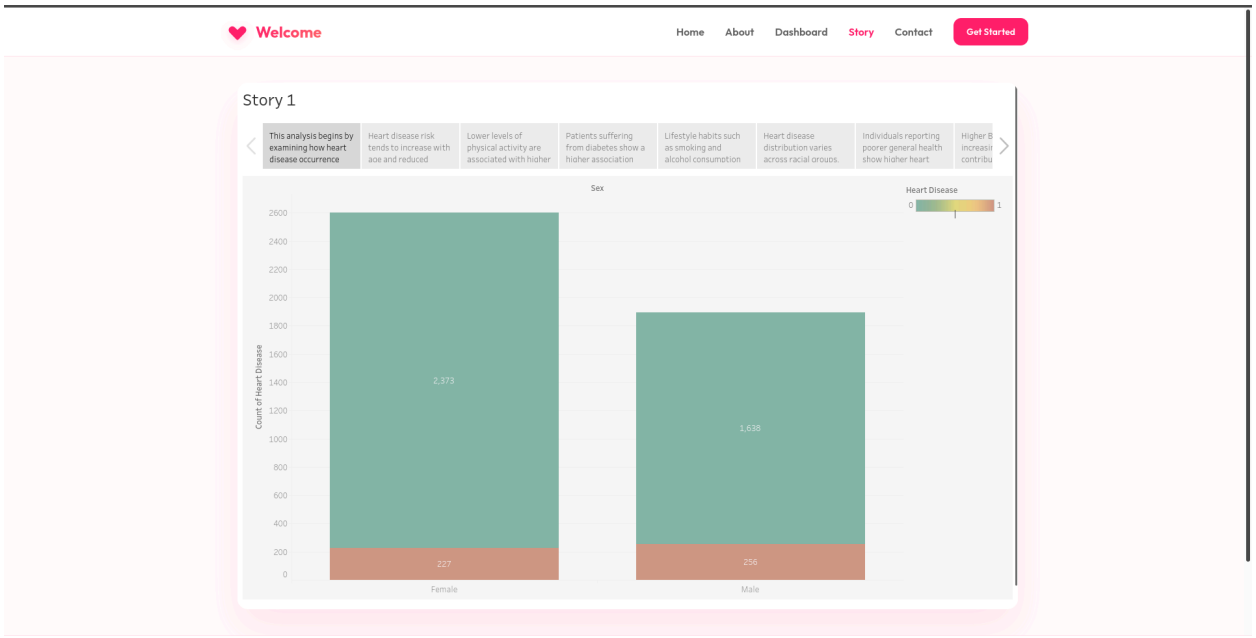
20. APPENDIX

- Github Link - https://github.com/kanak227/DA_HeartDisease
- Deployed web link - https://da-heartdisease.onrender.com/




Welcome
Home
About
Dashboard
Story
Contact
Get Started
Heart Disease Analysis Dashboard
Gender Vs Heart disease
Sex
Count of Heart Disease
2,373
1,638
227
256
Female
Male
Diabetic Vs Stroke
Heart Disease / Diabetic
Count of Stroke
No
No, border line..
Yes
Yes (during ..
Race
American Indian/Ala...
Asian
Black
Hispanic
Other
White
Heart Disease
Racewise Heart Disease
0.21%
17.81%
3.73%
75.57%
Impact of smoking and alcohol on heart disease
Smoking
Alcohol Drinking
Count of Heart Disease


Welcome
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Get Started
Story 1
This analysis begins by examining how heart disease occurrence
Heart disease risk tends to increase with age and reduced
Lower levels of physical activity are associated with higher
Patients suffering from diabetes show a higher association
Lifestyle habits such as smoking and alcohol consumption
Heart disease distribution varies across racial groups.
Individuals reporting poorer general health show higher heart
Sex
Heart Disease
Count of Heart Disease
2,373
1,638
227
256
Female
Male